

Day 5 – Shell Scripting: Functions & Modular Structure

Objective:

Learn how to use functions to build modular, reusable, and professional shell scripts.

1. Function Syntax:

```
function_name() {  
    commands  
}
```

2. Calling a Function:

```
function_name
```

3. Passing Arguments to Functions:

```
check_user() {  
    echo "Checking user: $1"  
}
```

4. Exit Status & Return Codes:

```
return 0 (success)
```

```
return 1 (failure)
```

```
$? holds last command exit status
```

5. Modular Script Structure:

```
validate_input() { ... }
```

```
check_service() { ... }
```

```
main() {
```

```
    validate_input "$@"
```

```
    check_service "$@"
```

```
}
```

```
main "$@"
```

Production Example:

```
check_service() {  
  for service in "$@"; do  
    if systemctl is-active --quiet "$service"; then  
      echo "$service is running"  
    else  
      echo "$service is NOT running"  
    fi  
  done  
}
```

Best Practices:

- Use modular functions
- Validate input early
- Quote variables
- Use exit codes for error handling
- Keep main() as entry point

Day 5 completed successfully